

HIROO, Japan

WMO: 474400

Lat: 42.2944N

Lon: 143.3167E

Elev: 34

StdP: 100.91

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.4	-10.7	-20.1	0.6	-9.1	-18.2	0.8	-8.2	10.1	-0.8	9.0	-0.9	2.3	270	0.543

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.3	(c) 27.7	(d) 21.7	(e) 25.6	(f) 20.3	(g) 23.4	(h) 19.1	(i) 22.6	(j) 26.5	(k) 21.5	(l) 24.4	(m) 20.3	(n) 22.4	(o) 4.3	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.3	16.0	24.7	20.3	15.1	23.3	19.5	14.3	22.0	67.2	26.7	62.9	24.3	58.8	22.5	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.2	(b) 7.7	(c) 6.5	(e) -15.2	(f) 31.5	(g) 1.9	(h) 2.0	(i) -16.5	(j) 33.0	(k) -17.6	(l) 34.1	(m) -18.7	(n) 35.2	(o) -20.0	(p) 36.6		
(a) 9.2	(b) 7.7	(c) 6.5	(e) -15.9	(f) 24.2	(g) 1.9	(h) 1.0	(i) -17.2	(j) 24.9	(k) -18.3	(l) 25.5	(m) -19.3	(n) 26.0	(o) -20.7	(p) 26.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-4.7	-4.3	-0.1	5.3	10.0	12.9	17.0	19.0	16.8	11.2	5.0	-1.5
	DBStd	8.77	3.03	3.18	3.37	3.52	3.86	3.24	3.22	3.07	2.91	3.04	3.75	3.30
	HDD10.0	1904	455	401	312	148	48	7	0	0	0	22	156	356
	HDD18.3	4128	713	634	571	392	258	166	65	28	65	222	401	614
	CDD10.0	914	0	0	0	7	49	94	217	279	204	58	5	0
	CDD18.3	97	0	0	0	1	1	4	24	48	18	1	0	0
	CDH23.3	477	0	0	0	1	18	28	144	217	68	1	0	0
	CDH26.7	107	0	0	0	0	3	4	34	56	9	0	0	0
Wind	WSAvg	2.8	3.1	3.1	3.2	3.2	2.8	2.2	2.0	2.2	2.5	2.9	3.2	3.2
Precipitation	PrecAvg	1564	65	55	75	106	151	132	164	200	242	161	124	89
	PrecMax	2086	152	139	130	369	285	398	389	485	482	344	308	225
	PrecMin	1078	9	9	17	14	52	25	25	53	63	48	33	27
	PrecStd	268	43	37	37	77	70	93	95	106	107	86	70	48
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.2	7.2	12.0	20.7	25.4	26.9	30.0	31.2	27.8	21.8	16.9	10.9
		MCWB	2.0	3.8	6.6	11.1	14.4	17.8	22.7	23.7	21.3	15.9	12.1	7.2
	2%	DB	3.4	4.7	9.0	16.6	21.6	23.2	27.2	28.1	25.3	19.4	14.9	7.6
		MCWB	0.6	1.4	4.8	9.3	13.1	16.6	21.7	22.4	20.2	14.3	10.5	4.6
	5%	DB	1.7	2.8	6.9	13.6	18.8	20.4	24.5	25.8	23.2	17.7	13.2	5.6
		MCWB	-1.0	-0.1	3.0	7.9	12.2	15.4	20.3	21.4	19.1	13.6	9.6	2.7
	10%	DB	0.5	1.2	5.1	11.1	16.3	17.6	21.6	23.3	21.3	16.3	11.4	3.8
		MCWB	-1.9	-1.2	1.8	6.3	11.1	14.6	18.9	20.3	18.2	12.8	8.2	1.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.0	4.5	7.7	12.3	15.9	19.2	23.6	24.4	23.0	17.6	13.3	7.8
		MCDB	4.9	6.4	11.2	18.0	22.0	24.8	28.6	29.9	25.9	19.6	15.7	10.3
	2%	WB	1.0	1.8	5.2	10.0	14.3	17.4	22.2	23.0	21.4	15.7	11.5	5.1
		MCDB	2.8	4.1	8.6	15.8	19.8	21.4	26.5	27.0	23.9	18.0	14.0	6.9
	5%	WB	-0.3	0.2	3.5	8.4	12.9	16.0	20.7	21.8	20.1	14.5	10.0	3.1
		MCDB	1.3	2.2	6.4	12.6	17.7	19.1	23.8	24.9	22.0	16.7	12.5	5.1
	10%	WB	-1.7	-1.0	2.2	6.9	11.6	14.7	19.1	20.8	18.7	13.5	8.3	1.4
		MCDB	0.3	1.0	4.6	10.4	15.4	17.0	21.1	22.9	20.6	15.7	11.0	3.6
Mean Daily Temperature Range	5% DB	MDBR	7.9	7.9	7.3	7.9	7.6	5.8	5.2	5.3	6.4	7.9	7.8	7.4
		MCDBR	8.7	9.4	9.7	12.8	13.3	11.7	10.8	9.9	9.2	9.3	10.2	8.9
		MCWBR	7.1	7.5	6.9	7.4	6.9	6.1	5.8	5.3	5.4	6.4	8.1	7.2
	5% WB	MCDBR	8.4	9.0	9.2	11.4	11.6	9.5	9.9	8.5	7.6	8.0	9.7	8.8
		MCWBR	7.4	7.6	7.1	7.2	6.7	5.5	5.8	5.0	5.0	6.1	8.3	7.5
Clear-Sky Solar Irradiance	taub	0.335	0.385	0.466	0.523	0.542	0.511	0.499	0.480	0.422	0.404	0.383	0.352	
	taud	2.167	2.011	1.866	1.811	1.855	2.022	2.128	2.197	2.316	2.225	2.183	2.156	
	Ebn at Noon	776	786	762	752	752	778	782	785	807	765	713	713	
	Edh at Noon	99	137	179	204	201	170	152	138	114	110	98	93	
All-Sky Solar Radiation	RadAvg	1.91	2.77	3.88	4.73	4.96	4.85	4.32	4.05	3.61	2.94	1.97	1.57	
	RadStd	0.10	0.18	0.30	0.42	0.47	0.42	0.42	0.37	0.32	0.19	0.14	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		+0.31	N/A	N/A	N/A	N/A	+0.83	N/A	-97	+73	N/A
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-63	N/A	N/A

Nomenclature: See separate page